

C Programming - HDOC Day 10

Programming Activities: Algorithm, Pseudocode, Flowchart, Test Cases, C Program, Compilation and Execution

Question 1: Time Conversion from Seconds

Aim: Write a C program to convert a given number of seconds into days, hours, minutes, and remaining seconds.

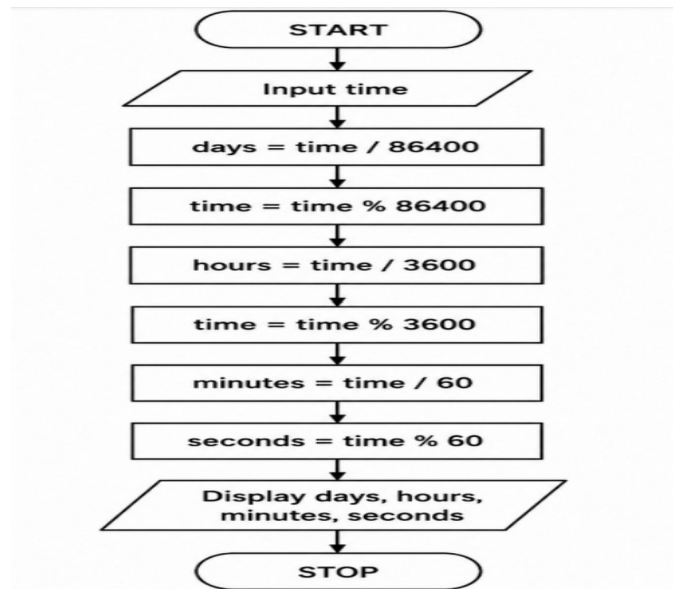
Algorithm

1. Start.
2. Input time in seconds.
3. Calculate $\text{days} = \text{time} / 86400$.
4. Keep the remaining seconds using $\text{time} = \text{time} \% 86400$.
5. Calculate $\text{hours} = \text{time} / 3600$.
6. Keep the remaining seconds using $\text{time} = \text{time} \% 3600$.
7. Calculate $\text{minutes} = \text{time} / 60$.
8. Calculate $\text{seconds} = \text{time} \% 60$.
9. Display days, hours, minutes and seconds.
10. Stop.

Pseudocode

```
START
INPUT time
days = time / 86400
time = time % 86400
hours = time / 3600
time = time % 3600
minutes = time / 60
seconds = time % 60
DISPLAY days, hours, minutes, seconds
STOP
```

Flowchart



Test Case Table

Test Case	Input (seconds)	Days	Hours	Minutes	Seconds
1	60	0	0	1	0
2	3661	0	1	1	1
3	86400	1	0	0	0
4	90061	1	1	1	1

C Program

```
#include <stdio.h>
```

```

int main()
{
    int time, days, hours, minutes, seconds;

    printf("Enter time in seconds: ");
    scanf("%d", &time);

    days = time / 86400;
    time = time % 86400;

    hours = time / 3600;
    time = time % 3600;

    minutes = time / 60;
    seconds = time % 60;

    printf("Days = %d\n", days);
    printf("Hours = %d\n", hours);
    printf("Minutes = %d\n", minutes);
    printf("Seconds = %d\n", seconds);

    return 0;
}

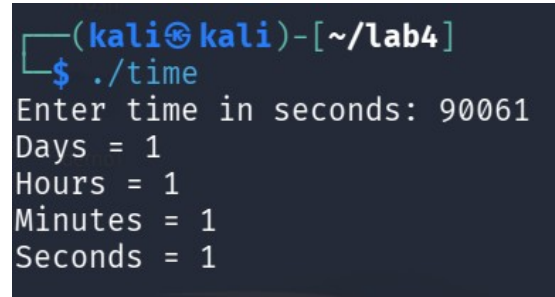
```

Compilation and Execution

```
gcc time.c -o time
./time
```



```
(kali@kali)-[~/lab4]
$ gcc time.c -o time
```



```
(kali@kali)-[~/lab4]
$ ./time
Enter time in seconds: 90061
Days = 1
Hours = 1
Minutes = 1
Seconds = 1
```

Result

The program executed successfully and converted the supplied time value into days, hours, minutes, and seconds. The test cases verified that the calculated values were correct.

Question 2: Leap Year Verification

Aim: Design a C program that accepts a year and checks whether it satisfies the conditions of a leap year.

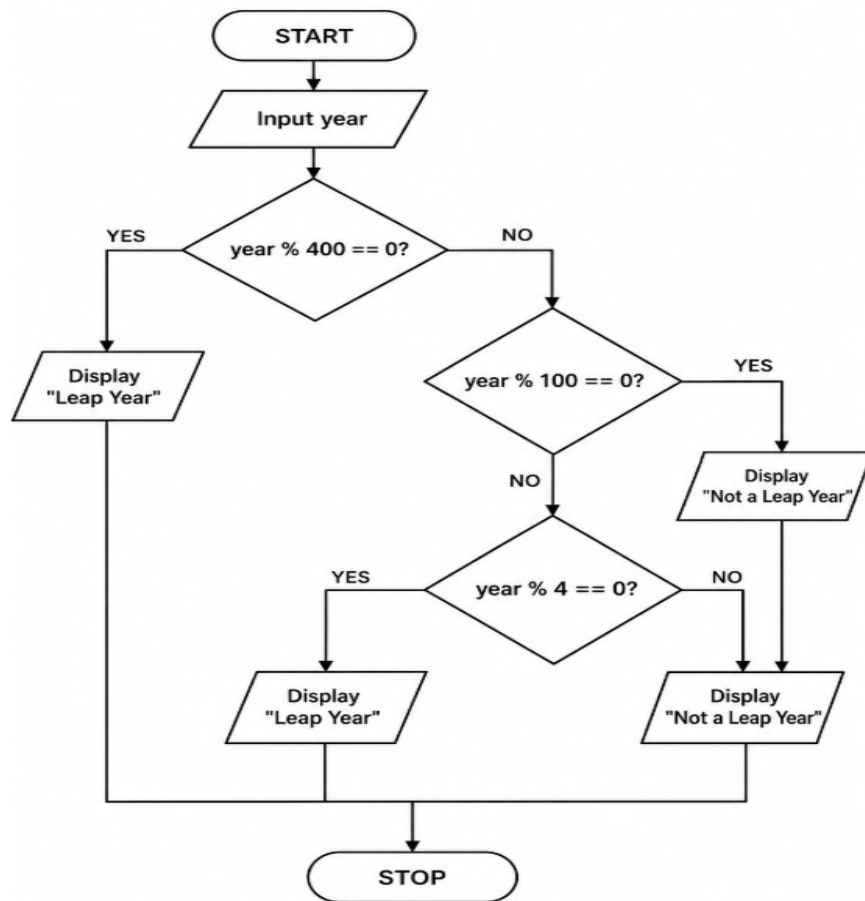
Algorithm

11. Start.
12. Input a year.
13. If the year is divisible by 400, display "Leap Year".
14. Else if the year is divisible by 100, display "Not a Leap Year".
15. Else if the year is divisible by 4, display "Leap Year".
16. Otherwise, display "Not a Leap Year".
17. Stop.

Pseudocode

```
START
INPUT year
IF year % 400 == 0
    DISPLAY "Leap Year"
ELSE IF year % 100 == 0
    DISPLAY "Not a Leap Year"
ELSE IF year % 4 == 0
    DISPLAY "Leap Year"
ELSE
    DISPLAY "Not a Leap Year"
STOP
```

Flowchart



Test Case Table

Test Case	Input Year	Expected Output
1	2024	Leap Year
2	2025	Not a Leap Year
3	1900	Not a Leap Year
4	2000	Leap Year

C Program

```
#include <stdio.h>
```

```
int main()
{
    int year;

    printf("Enter year: ");
    scanf("%d", &year);

    if (year % 400 == 0)
        printf("Leap Year");
    else if (year % 100 == 0)
        printf("Not a Leap Year");
    else if (year % 4 == 0)
        printf("Leap Year");
    else
        printf("Not a Leap Year");
}
```

```
    return 0;  
}
```

Compilation and Execution

```
gcc leapyear.c -o leapyear  
./leapyear
```

```
(kali㉿kali)-[~/lab4]  
$ gcc leapyear.c -o leapyear
```

```
(kali㉿kali)-[~/lab4]  
$ ./leapyear  
Enter year: 2024  
Leap Year
```

```
(kali㉿kali)-[~/lab4]  
$ ./leapyear  
Enter year: 2025  
Not a Leap Year
```

<pre>(kali㉿kali)-[~/lab4] \$./leapyear Enter year: 1900 Not a Leap Year</pre>	<pre>(kali㉿kali)-[~/lab4] \$./leapyear Enter year: 2000 Leap Year</pre>
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Result:

The C program was successfully compiled and executed. It correctly determines whether the entered year is a **leap year or not**. The program was tested using the prepared test cases and produced the expected output.